-the first step in text analysis is deciding what to mine, it can be collected from emails, instant messages and blogs, but the most common source is news.We then have to remove unnecessary information from the collected data, like ads or images. It's important to keep meta-data.

-the next step is to correct sentence boundary by detecting sentence units such as sentences ending with "!"

-We then have to extract meaningful tokens, the task is challenging because you have to determine if you have to use one or two tokens depending on the sentence.

-It's important to understand that the word with the highest frequency doesn't mean that the word is the most important. Stop words do not carry information, removing them helps machine learning algorithms to perform better.

-Stemming is the process of reducing words to their base form, helping normalize words with same meaning but different spelling. Lemmatization is the process to find the actual root of a word.

-Text transformation represents documents using methods like bag-of-words or TF-IDF, which weighs words based on their frequency in a document versus the number of documents containing the word. A way to represent a sentence is the one-hot-vector representation, where each word is represented as a vector, where the corresponding word is assigned to 1 and other position to 0.

-word embedding creates a visual representation of semantic representation between words using vector spaces. Public pre-trained word models exist, such as Word2Vec, GloVe and FastText.

-The next step is selection and extraction of the best features. Feature extraction can be done through Latent Semantic Analysis based on singular value decomposition.

-Latent Semantic Analysis (LSA) is a feature extraction technique that uses singular value decomposition to reduce the dimensionality of text data while preserving its essential information. We can visualize the similarities and differences between research toppings using 2D spaces.

-Toping modeling like Doc2Vec can work as feature extractor, using figures to predict the next words based on the previous ones.

-When calculating the similarity between documents, cosine similarity is preferred. It measures similarity using the angle between the vectors of the document.

-Machine learning is also used for classification: such as spam filtering using vector space model, domain knowledge-based phrases and meta-data information.

-Document clustering is used to see relationships between toppings and keyword association.

-information retrieval is used to find useful information from databases or by using topic modeling. LDA can discover toppings within text data and can analyses similarity between documents.